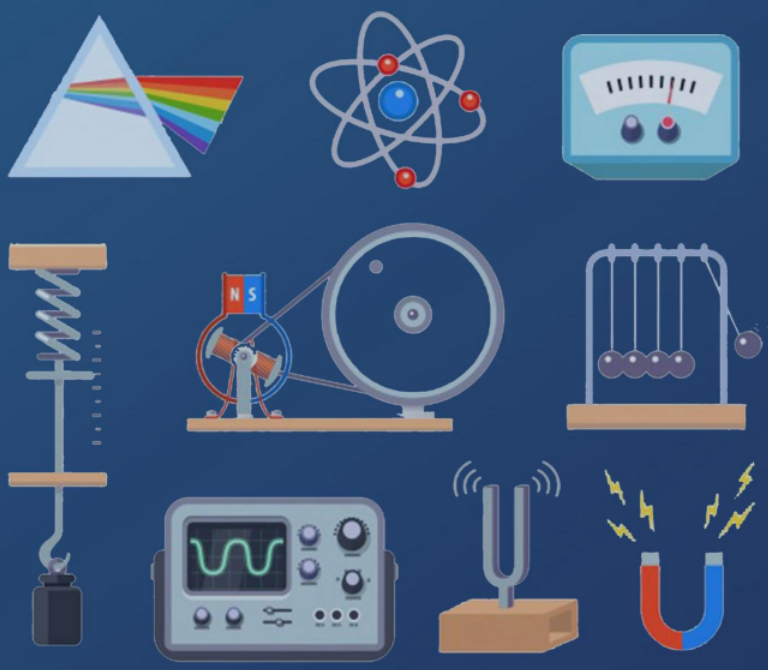
PHYSICS

Magnetic Effect of Current



SHEET

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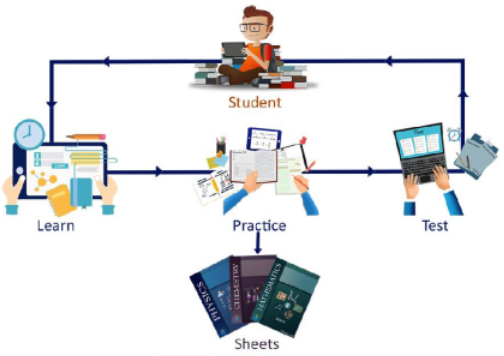
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Biot - Savart Law	1			
2	Magnetic field due to straight finite current carrying wire	2		1-4	
3	Magnetic field Lines due to straight infinite current carrying wire	3-5	1,2	5-10	1
4	Magnetic field due to straight current carrying wire	6	3,4		
5	Magnetic field due to circular current loop	7-10	14	11	
6	Ampere's circuital law		10	12	
7	Magnetic field due to long cylinders		5	13-16	2,3,18-20
8	Solenoid and Toroid	11-13	15		
9	Magnetic force on a moving charged particle	14-17		17-20	
10	Motion of particle in magnetic field	18-21	6,11	21-29	4-6
11	Helical path		13	30	7,8,21-24, 33
12	Motion of particle in combined electric and magnetic field	22-24	7	31	9,10, 25-28
13	Cyclotron			32.33	
14	Hall effect			34	
15	Magnetic force on a current carrying wire	25,26		35-38	11
16	Force between two parallel and perpendicular currents	27-30			29-32
17	Torque on a current loop	31,32	8,9	39-43	12,34
18	Circular current loop as a magnetic dipole	33	16		
19	Moving coil galvanometer				
20	After full Chapter		12		13-17

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Biot - Savart Law

Q1. The vector form of Biot savart law for a current carrying element is :–

(A) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{id\ell \sin\phi}{r^2}$

(B) $d\vec{B} = \frac{\mu_0}{4\pi}\frac{i\vec{d\ell} \times \hat{r}}{r^2}$

(C) $\vec{B} = \frac{\mu_0}{4\pi}\frac{i\vec{d\ell} \times \hat{r}}{r^3}$

(D) $\vec{B} = \frac{\mu_0}{4\pi}\frac{i\vec{d\ell} \times \vec{r}}{r^2}$

Magnetic field due to straight finite current carrying wire

Q2. A vertical straight conductor carries a current vertically upwards. A point P lies to the east of it at a small distance and another point Q lies to the west at the same distance. The magnetic field at P is :-

(A) Greater than at Q

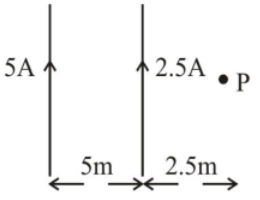
(B) Same as at Q

(C) Less than at Q

(D) Greater or less than at Q, depending upon the strength of current

Magnetic field due to straight infinite current carrying wire

Q3. Magnetic field at point 'P' due to both infinite long current carrying wires is :-



(A) $\frac{\mu_0}{2\pi}\otimes$ (B) $\frac{5\mu_0}{6\pi}\otimes$

(C) $\frac{5\mu_0}{6\pi}\odot$ (D) $\frac{\mu_0}{2\pi}\odot$

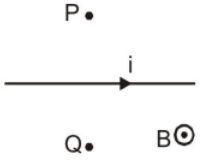
Q4. A long, straight wire carrying a current of 1.0 A is placed horizontally in a uniform magnetic field B = 1.0×10^{-5} T pointing vertically upward (figure). The magnitude of the resultant magnetic field at the points P and Q, both situated at a distance of 2.0 cm from the wire in the same horizontal plane are respectively

(A) zero, 20 μT

(B) 20 μT, zero

(C) zero, zero

(D) 20 μT, 20 μT



Q5. A current of 10 A is established in a long wire along positive z -direction. The magnetic field B at the point (1m, 0, 0) is :-

(A) 1 μT along the –y direction

(B) 2μT along the +y direction

(C) 1μT along the –x direction

(D) 2μT along the +x direction

Magnetic field Line due to straight current carrying wire

Q6. Due to a finitely long current carrying wire, magnetic field lines will be-

(A) Straight lines

(B) Concentric circles with centres on the wire

(C) Both A snd B

(D) None of these

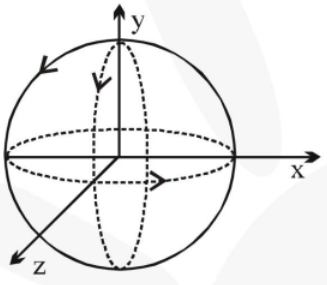
Magnetic field due to circular current loop

Q7. A circular coil of one turn is formed by a 6.28m length wire, which carries a current of 3.14A. The magnetic field at the centre of coil is :-

(A) 1×10^{-6} T (B) 4×10^{-6} T

(C) 0.5×10^{-6} T (D) 2×10^{-6} T

Q8. Three rings, each having equal radius R, are placed mutually perpendicular to each other and each having its centre at the origin of co-ordinate system. If current I is flowing through each ring then the magnitude of the magnetic field at the common centre is -



(A) $\sqrt{3}\frac{\mu_0 I}{2R}$ (B) zero

(C) $\left(\sqrt{2}-1\right)\frac{\mu_0 I}{2R}$ (D) $\left(\sqrt{3}-\sqrt{2}\right)\frac{\mu_0 I}{2R}$

Q9. Two concentric coils X and Y of radii 16 cm and 10 cm lie in the same vertical plane containing N-S direction. X has 20 turns and carries 16 A. Y has 25 turns & carries 18A. X has current in anticlockwise direction and Y has current in clockwise direction for an observer, looking at the coils facing the west. The magnitude of net magnetic field at their common centre is -

(A) $5\pi \times 10^{-4}$ T towards west

(B) $13\pi \times 10^{-4}$ T towards east

(C) $13\pi \times 10^{-4}$ T towards west

(D) $5\pi \times 10^{-4}$ T towards east

Q10. A length of wire carries a steady current, is bent first to form a plane circular coil of one turn, same length now bent more sharply to give three turns of smaller radius. Magnetic field becomes:

(A) 3 times (B) 1/3 times

(C) 9 times (D) unchange

Solenoid and Toroid

Q11. A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A. estimate the magnitude of B inside the solenoid near its centre.

(A) zero (B) $8\pi \times 10^{-3}$ T

(C) $15\pi \times 10^{-3}$ T (D) $\pi \times 10^{-3}$ T

Q12. If number of turns and current becomes double for any solenoid, then value of magnetic field becomes :-

(A) twice (B) same

(C) half (D) four times

Q13. Mean radius of a toroid is 10 cm and number of turns are 500. If current flowing through it is 0.1 ampere then value of magnetic field (in tesla) for toroid :-

(A) 10^{-2} (B) 10^{-5}

(C) 10^{-3} (D) 10^{-4}

Magnetic force on a moving charged particle

Q14. A test charge $1{\cdot}6 \times 10^{-19}$ C is moving with velocity $\vec{v} = (2\hat{i} + 3\hat{j})$ m/sec in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j})$ Wb/m^2. The magnetic force on the test charge :-

(A) $6\hat{k}$T

(B) $\left(4\hat{i}+6\hat{j}\right)$T

(C) $\left(4\hat{i}+6\hat{j}\right)\times 10^{-19}$T

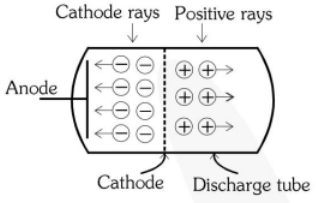
(D) zero

Q15. In a region a uniform magnetic field acts in horizontal plane towards north. If cosmic particles (80% protons) falling vertically downwards, then they are deflected towards-

(A) North (B) South

(C) East (D) West

Q16. A magnetic field is applied at perpendicular to the axis of cylinderical discharge tube then cathode and positive rays deflected in -



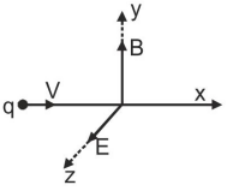
(A) Same direction.

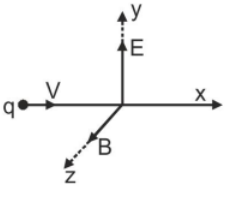
(B) Opposite direction.

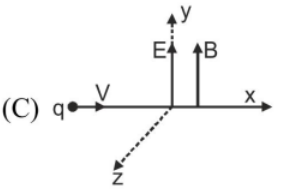
(C) Mutually perpendicular direction.

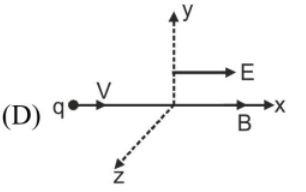
(D) No deflection for both.

Q17. A particle of charge q and mass m is moving along the x-axis with a velocity v and enters a region of electric field E and magnetic field B as shown in figure beolw. For which figure the net force on the charge may be zero :-

(A)


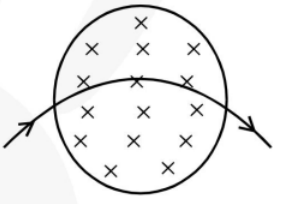
(B)


(C)


(D)


Motion of particle in magnetic field

Q18. There is a magnetic field acting in a plane perpendicular downwards. A particle in vacuum moves in the plane of paper from left to right as shown in figure. The path indicated by the arrow could be due to -



(A) Proton (B) Neutron

(C) Electron (D) Alpha particle

Q19. **Statement–1** : A magnetic field may interacts with a moving charge and not with a stationary charge.

Statement–2 : A moving charge produces a magnetic field.

(A) Statement–1 is True, Statement–2 is False.

(B) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is True ; Statement–2 is NOT a correct explanation for Statement–1.

(D) Statement–1 is False, Statement–2 is True.

Q20. An electron makes 3×10^5 revolutions per second in a circle of radius 0.5 angstrom. Find the magnetic field B at the centre of the circle.

(A) 6×10^{-10} T (B) 12×10^{-10} T

(C) 18×10^{-10} T (D) 24×10^{-10} T

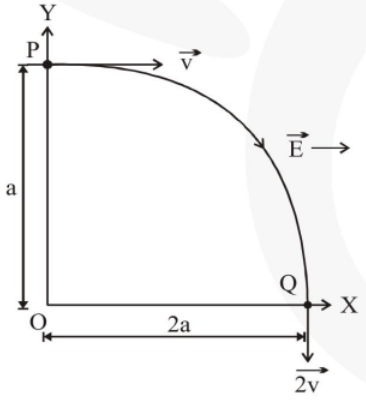
Q21. Which of the following charge has maximum frequency of rotation in uniform transverse magnetic field :-

(A) a proton (B) an alpha particle

(C) an electron (D) a neutron

Motion of particle in combined electric and magnetic field

Q22. A particle of charge +q and mass m moving under the influnce of a uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows trajectory from P to Q as shown in figure. The velocities at P and Q $v\hat{i}$ and $-2v\hat{j}$ respectively. Which of the following statement is incorrect -



(A) $E = \frac{3}{4}\left[\frac{mv^2}{qa}\right]$

(B) Rate of work done by electric field at P is $\frac{3}{4}\left[\frac{mv^3}{a}\right]$

(C) Rate of work done by electric field at P is zero

(D) Rate of work done by both the fields at Q is zero

Q23. Electric field strength $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{i}$ exists in a region. A charge is projected with a velocity $\vec{v} = v_0\hat{j}$ at origin, then -

(A) It moves along helix with constant pitch

(B) It moves along circular path in YZ plane

(C) It moves along helix with increasing pitch

(D) It moves along helix with decreasing pithch

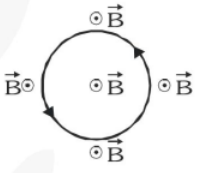
Q24. A charged particle is released from rest in a region of steady and uniform electric and magnetic fields which are parallel to each other. The particle will move in a :

(A) straight line (B) circle

(C) helix (D) cycloid

Magnetic force on a current carrying wire

Q25. An elastic circular wire of length ℓ carries a current I_0 It is placed in a uniform magnetic field $\vec{B}$ (out of paper) such that its plane is perpendicular to the direction of $\vec{B}$. The wire will experience:



(A) no force

(B) a stretching force

(C) a compressive force

(D) a torque

Q26. A conductor in the form of a right angle ABC, with AB=3cm and BC = 4cm, carries a current of 10A. There is a uniform magnetic field of 5T perpendicular to the plane of the conductor. The force on the conductor will be :-

(A) 1.5N (B) 2.0N

(C) 2.5N (D) 3.5N

Force between two parallel and perpendicular currents

Q27. **Statement–1 :** A solenoid tends to expand, when a current passes through it.

Statement–2 : Two straight parallel metallic wires carrying current in opposite direction attract each other.

(A) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.

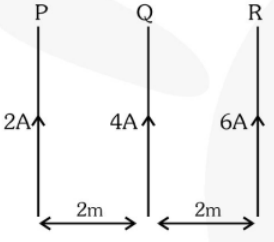
(B) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1.

(C) Statement–1 is True, Statement–2 is False.

(D) Both Statement–1 and Statement-2 are False.

Q28. Three long straight wires, carries currents are arranged according to figure.

Magnetic force on 10cm part of wire Q is:



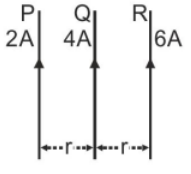
(A) 16×10^{-9} N, towards right

(B) 16×10^{-8} N, towards right

(C) 16×10^{-8} N, towards left

(D) 16×10^{-9} N, towards left

Q29. P, Q and R long parallel straight wires in air, carrying currents as shown. The direction of resultant force on R is :-



(A) towards left

(B) towards right

(C) the same as that of current in Q

(D) perpendicular to plane of paper

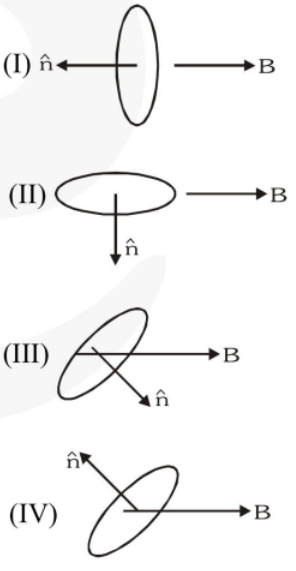
Q30. Three long straight conductors are arranged parallel to each other in the same plane and carry currents of 1 A, 2 A and 3 A all in the same direction. The distance between the first two conductors is "x" and the distance between the second and third conductors is "y". If the middle conductor is in equilibrium, the ratio x : y is

(A) 1 : 3 (B) 3 : 1

(C) 1 : $\sqrt{3}$ (D) $\sqrt{3}$: 1

Torque on a current loop

Q31. A current carrying loop is placed in a uniform magnetic field in four different orientations, I, II, III, IV, arrange them in the decreasing order of potential energy :



(A) I > III > II > IV

(B) I > II > III > IV

(C) I > IV > II > III

(D) III > IV > I > II

Q32. Two particles, each of mass m and charge q, are attached to the two ends of a light rigid rod of length 2R. The rod is rotated at constant angular speed about a perpendicular axis passing through its centre. The ratio of the magnitudes of the magnetic moments of the system and its angular momentum about the centre of the rod is :

(A) q/2m (B) q/m

(C) 2q/m (D) q/πm

Circular Current Loop as a Magnetic Dipole

Q33. A circular loop has a radius of 5 cm. and it is carrying a current of 0.1 Amp. its magnetic moment is :–

(A) 1.32×10^{-4} amp × m^2

(B) 2.62×10^{-4} amp × m^2

(C) 5.25×10^{-4} amp × m^2

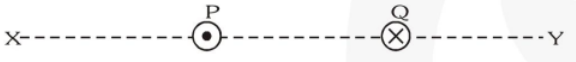
(D) 7.85×10^{-4} amp × m^2

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two long and parallel wires at P and Q carry currents 0.1 A and 0.2 A respectively in opposite directions as shown.



In which region(s) a neutral point may not exist, that is where the magnetic field is not zero?

(A) Between X and P

(B) Between P and Q

(C) Between Q and Y

(D) In all the regions there exists no neutral point

Q2. A long straight wire carries a current along the x-axis. Consider the points A(0, 1, 0), B(0, 1, 1), C(1, 0, 1) and D(1, 1, 1). Which of the following pairs of points will have magnetic fields of the same magnitude?

(A) A and B (B) A and C

(C) B and C (D) B and D

Q3. Consider the magnetic field produced by a finitely long current carrying wire

(A) The lines of field will be concentric circles with centres on the wire.

(B) There can be two points in the same plane where magnitude of magnetic fields are same.

(C) There can be large number of points where magnitude of magnetic field is same.

(D) The magnetic field at a point is inversely proportional to the distance of the point from the wire.

Q4. The magnetic lines of force due to a straight current carrying wire will be:

(A) circular for finite length of wire

(B) circular for semi-infinite wire

(C) circular for infinite wire

(D) Parabolic for finite wire

Q5. A current I flows along the length of an infinitely long, straight, solid pipe. Then correct statement(s)-

(A) The magnetic field is zero only on the axis of the pipe.

(B) The magnetic field is different at different points inside the pipe.

(C) The magnetic field is maximum on surface.

(D) The magnetic field at all points inside the pipe is the same, but not zero.

Q6. An electron is moving along the positive X-axis. You want to apply a magnetic field for a short time so that the electron may reverse its direction and move parallel to the negative X–axis. This can be done by applying the magnetic field along :-

(A) Y-axis

(B) Z-axis

(C) Y-axis only

(D) Z-axis only

Q7. If a charge projected in a zero gravity region. Find possible cases of electric and magnetic field in that region so no net force exerts on charge.

(A) $E = 0, B = 0$

(B) $E = 0, B \neq 0$

(C) $E \neq 0, B \neq 0$

(D) $E \neq 0, B = 0$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 8 & 9

Torque acting on a current loop in uniform magnetic field is given by $\vec{\tau} = \vec{M} \times \vec{B}$. But force on it is zero. If it is free to rotate then it will rotate about an axis passing through its centre of mass and parallel to $\vec{\tau}$. The potential energy of loop is given by $U = -\vec{M}.\vec{B}$. A current carrying ring with its centre at origin and moment of inertia 10^{-2} kg–m^2 about an axis passing through its centre and perpendicular to its plane has magnetic moment $\vec{M} = (3\hat{i} - 4\hat{j})$ A–m^2. At time $t = 0$ a magnetic field $\vec{B} = (4\hat{i} - 3\hat{j})$ T is switched on.

Q8. Angular acceleration of the ring at time $t = 0$ in rad/s^2 is -

(A) 5000

(B) 1250

(C) 2500

(D) zero

Q9. Maximum angular velocity of the ring in rad/s will be :

(A) $50\sqrt{2}$

(B) $25\sqrt{2}$

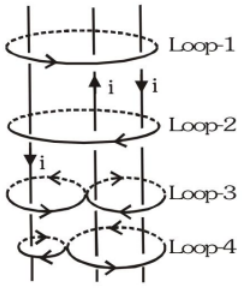
(C) $100\sqrt{2}$

(D) $150\sqrt{2}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10. Three wires are carrying same constant current i in different directions. Four loops enclosing the wires in different manners are shown. The direction of $d\vec{\ell}$ is shown in figure.



Column I	Column II
(A) Along closed loop–1	(p) $\oint \vec{B}.d\vec{\ell} = \mu_0 i$
(B) Along closed loop–2	(q) $\oint \vec{B}.d\vec{\ell} = -\mu_0 i$
(C) Along closed loop–3	(r) $\oint \vec{B}.d\vec{\ell} = 0$
(D) Along closed loop–4	(s) Net work done by the magnetic force to move a unit charge along the loop is zero.

Q11. A charged particle is moving in a circular path in uniform magnetic field. Match the following:

Table–I	Table–II
(A) Equivalent current due to motion of charge particle	(p) is proportional to v
(B) Magnetic moment	(q) is proportional to v^2
(C) Magnetic field at centre of circle due to motion of charged Particle.	(r) is proportional to v^0
	(s) None

Q12.

Column I	Column II
(A) A charge at rest produces	(p) Magnetic field
(B) A charge moving with uniform velocity produces	(q) Electric field
(C) An accelerated charge produces	(r) Electromagnetic waves

NUMERICAL TYPE

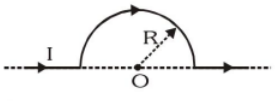
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q13. A beam of proton moving with velocity 4×10^5 m/sec enters in a uniform magnetic field of 0.3 T at an angle of 60° to the magnetic field. Calculate radius of helical path and pitch of helix.

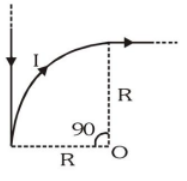
SUBJECTIVE TYPE

Q14. Find out magnetic field at point 'O' for the following current distributions.

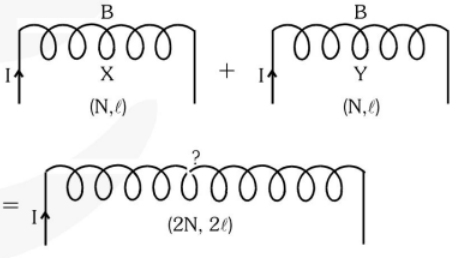
(a)



(b)



Q15. Two identical long solenoids X and Y carries equal current. The magnetic field produced at their axial mid point is B. If both the solenoids joined to each other according to figure. Find magnetic field at axial mid point of new solenoid.

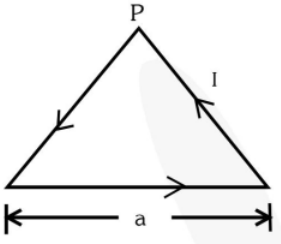


Q16. Two identical current carrying coils are concentric and perpendicular to each other. If magnetic moment of each coil is 'M'. Find resultant magnetic moment of two coils.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

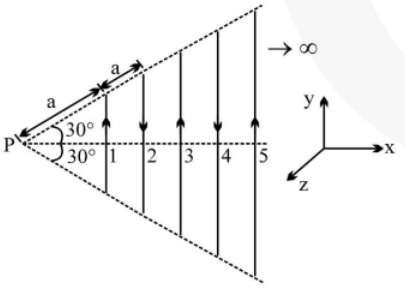
Q1. An equilateral triangle of side 'a' carries a current 'I'. Magnetic field at point 'P' which is vertex of triangle. :-



(A) $\frac{\mu_0 I}{2\sqrt{3}\pi a}$ □ (B) $\frac{\mu_0 I}{2\sqrt{3}\pi a}\otimes$

(C) $\frac{9}{2}\left(\frac{\mu_0 I}{\pi a}\right)$ □ (D) $\frac{9}{2}\left(\frac{\mu_0 I}{\pi a}\right)\otimes$

Q2. Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is -



(A) $\frac{\mu_0 I}{4\pi}\frac{\ln 2}{\sqrt{3}\,a}\hat{k}$ (B) $\frac{\mu_0 I}{4\pi}\frac{\ln 4}{\sqrt{3}\,a}\hat{k}$

(C) $\frac{\mu_0 I}{4\pi}\frac{\ln 4}{\sqrt{3}\,a}(-\hat{k})$ (D) Zero

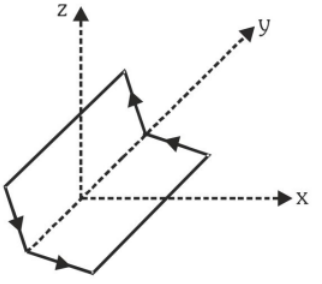
Q3. A non–planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length 2a. The magnetic field due to this loop at the point P (a, 0, a) points in the direction :

(A) $\frac{1}{\sqrt{2}}\left(-\hat{j}+\hat{k}\right)$

(B) $\frac{1}{\sqrt{3}}\left(-\hat{j}+\hat{k}+\hat{i}\right)$

(C) $\frac{1}{\sqrt{3}}\left(\hat{i}+\hat{j}+\hat{k}\right)$

(D) $\frac{1}{\sqrt{2}}\left(\hat{i}+\hat{k}\right)$



Q4. Consider a 10 cm long piece of a wire which carries a current of 10 A. Find the magnitude of the magnetic field due to the piece at a point which makes an equilateral triangle with the ends of the piece.

(A) Zero (B) 5.5 μT

(C) 7.5 μT (D) 11.5 μT

Q5. A straight wire carries current in vertical upward direction. A point 'P' lies just east of it at a small distance and another point 'Q' lies to the west just same distance. The magnetic field at these points :-

(consider earth magnetic field)

(A) $B_P > B_Q$ (B) $B_P < B_Q$

(C) $B_P = B_Q$ (D) None

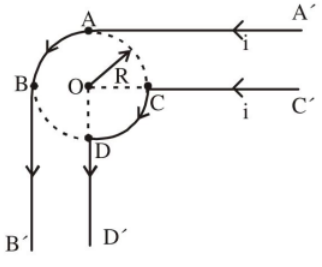
Q6. All straight wires are very long. Both AB and CD are arc of the same circle, both subtending right angles at the centre O. Then the magnetic field at O is

(A) $\frac{\mu_0 i}{4\pi R}$

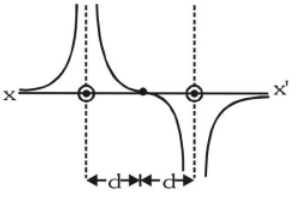
(B) $\frac{\mu_0 i}{4\pi R}\sqrt{2}$

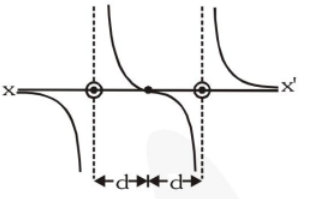
(C) $\frac{\mu_0 i}{2\pi R}$

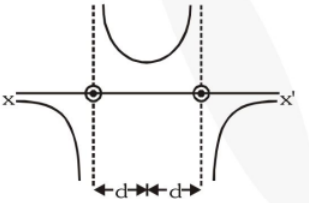
(D) $\frac{\mu_0 i}{2\pi R}(\pi+1)$

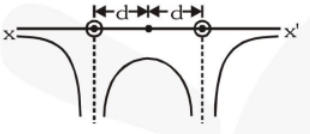


Q7. Two long parallel wires are at a distance 2d apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by :

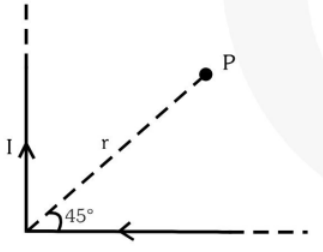
(A) 

(B) 

(C) 

(D) 

Q8. Magnetic field at point 'P' due to given current distribution is :-



(A) $\frac{\mu_0 I}{4\pi r}\left(1+\sqrt{2}\right)\odot$

(B) $\frac{\mu_0 I}{2\pi r}\left(1+\sqrt{2}\right)\odot$

(C) $\frac{\mu_0 I}{4\pi r}\left(1+\sqrt{2}\right)\otimes$

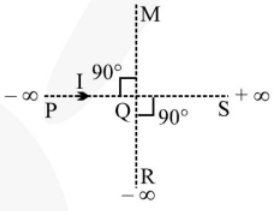
(D) $\frac{\mu_0 I}{2\pi r}\left(1+\sqrt{2}\right)\otimes$

Q9. A long straight wire along the z-axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the z = 0 plane is

(A) $\frac{\mu_0 I}{2\pi}\frac{(y\hat{i}-x\hat{j})}{(x^2+y^2)}$ (B) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{i}+y\hat{j})}{(x^2+y^2)}$

(C) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{j}-y\hat{i})}{(x^2+y^2)}$ (D) $\frac{\mu_0 I}{2\pi}\frac{(x\hat{i}-y\hat{j})}{(x^2+y^2)}$

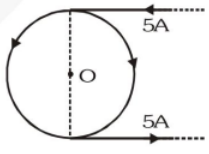
Q10. An infinitely long conductor PQR is bent to form a right angle as shown. A current I flows through PQR. The magnetic field due to this current at the point M is H_1. Now, another infinitely long straight conductor QS is connected at Q so that the current in PQ remaining unchanged. The magnetic field at M is now H_2. The ratio H_1/H_2 is given by:



(A) 1/2 (B) 1

(C) 2/3 (D) 2

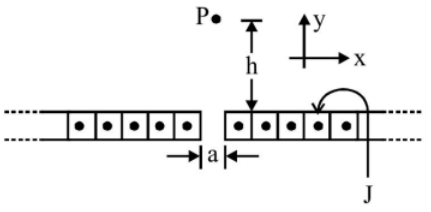
Q11. Magnetic field at point 'O' due to given current distribution. If 5A current is flowing in this system and the diameter of the loop is 10cm.



(A) 2×10^{-5} T, $\otimes$ (B) 10^{-5}T, $\odot$

(X) 10^{-5} T, $\odot$ (D) 2×10^{-5} T, $\odot$

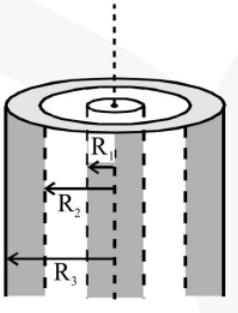
Q12. A small segment of length a is cut along z-axis from a infinite sheet having a surface current density J (current per unit width). The magnetic field at P is :-



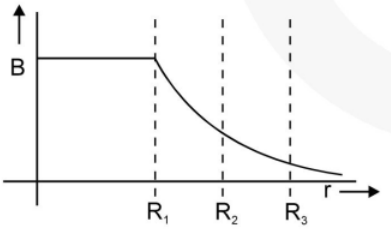
(A) $2\mu_0 J\left(1-\frac{h}{a\pi}\right)\hat{i}$ (B) $\frac{\mu_0 Jh}{2a\pi}\hat{i}$

(C) $\frac{\mu_0 J}{2}\left(\frac{a}{h\pi}-1\right)\hat{i}$ (D) $-\frac{\mu_0 j}{2}\left(\frac{h}{a\pi}-1\right)\hat{i}$

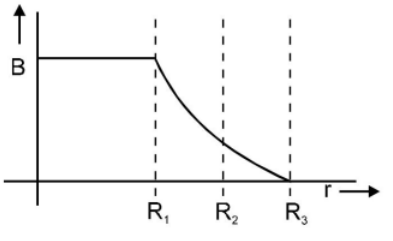
Q13. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius R_1 & the outer conductor is hollow of inner radius R_2 and outer radius R_3. The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as:



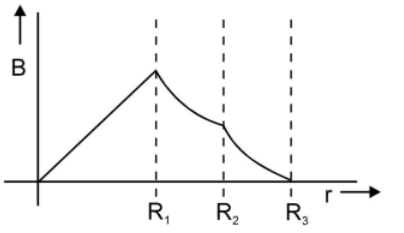
(A)



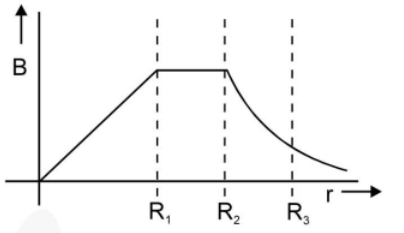
(B)



(C)



(D)



Q14. A long, cylindrical wire of radius b carries a current i distributed uniformly over its cross-section. Find the magnitude of the magnetic field at a point inside the wire at a distance a from the axis.

(A) zero (B) $\frac{\mu_0 ib}{2\pi a^2}$

(C) $\frac{\mu_0 ia^2}{2\pi b^3}$ (D) $\frac{\mu_0 ia}{2\pi b^2}$

Q15. A solid cylinderical wire of radius 'R' carries a current 'I'. The magnetic field is 5μT at a point, which is '2R' distance away from the axis of wire. Magnetic field at a point which is R/3 distance inside from the surface of the wire is :-

(A) $\frac{10}{3}\mu T$ (B) $\frac{20}{3}\mu T$

(C) $\frac{5}{3}\mu T$ (D) $\frac{40}{3}\mu T$

Q16. A hollow cylinderical wire carries a current I, having inner and outer radii 'R' and 2R respectively. Magnetic field at a point which 3R/2 distance away from its axis is :-

(A) $\frac{5\mu_0 I}{18\pi R}$ (B) $\frac{\mu_0 I}{36\pi R}$

(C) $\frac{5\mu_0 I}{36\pi R}$ (D) $\frac{5\mu_0 I}{9\pi R}$

Q17. A electron experiences a force $\left(4.0\hat{i}+3.0\hat{j}\right) \times 10^{-13}$ N in a uniform magnetic field when its velocity is $2.5\hat{k}\times10^{7}\,ms^{-1}$. When the velocity is redirected and becomes $\left(1.5\hat{i}-2.0\hat{j}\right)\times10^{7}\,ms^{-1}$, the magnetic force on the electron is zero. The magnetic field vector $\vec{B}$ is :

(A) $-0.075\hat{i}+0.1\hat{j}$ (B) $0.1\hat{i}+0.075\hat{j}$

(C) $0.075\hat{i}-0.1\hat{j}+\hat{k}$ (D) $0.075\hat{i}-0.1\hat{j}$

Q18. An electron moving with a velocity $\vec{V}_1 = 2\hat{i}$ m/s at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}N$. If the electron is moving with a velocity $\vec{V}_2 = 2\hat{j}$ m/s at the same point, it experiences a force $\vec{F}_2 = +2\hat{i}\,N$. The force the electron would experience if it were moving with a velocity $\vec{V}_3 = 2\hat{k}$ m/s at the same point is -

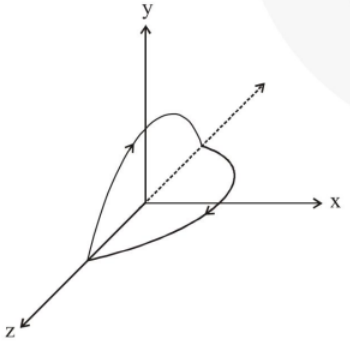
(A) zero

(B) $2\hat{k}N$

(C) $-2\hat{k}N$

(D) information is insufficient

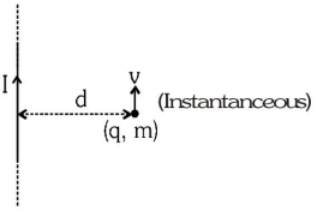
Q19. A non-planar circular loop consists of two semi-circles one of which lies in yz-plane & the other is in xz-plane as shown . The magnetic force experienced by positive charge of value Q moving with velocity v along x direction when it is at the origin is :



(A) $\frac{Qv\mu_0 I}{4R}$ (B) $\frac{Qv\mu_0 I}{2R}$

(C) $\frac{Qv\mu_0 I}{2\sqrt{2}R}$ (D) 0

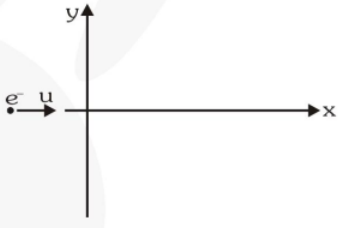
Q20. An electron is moving at 10^6 m/sec in a direction parallel to a current of 5A flowing through an infinite long straight wire separated by a perpendicular distance of 10cm in air. Magnetic force experienced by the electron :-



(A) 1.6×10^{-19} N (B) 1.6×10^{-20} N

(C) 1.6×10^{-18} N (D) 1.6×10^{-21} N

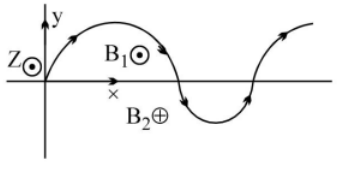
Q21. An electron moving with a speed u along the positive x–axis at y = 0 enters a region of uniform magnetic field $\vec{B} = -B_0\hat{k}$ which exists to the right of y–axis. The electron exits from the region after sometime with the speed v at co–ordinate y, then :



(A) $v > u,\ y < 0$ (B) $v = u,\ y > 0$

(C) $v > u,\ y > 0$ (D) $v = u,\ y < 0$

Q22. At t = 0 a charge q is at the origin and moving in the y-direction with velocity $\vec{v} = v\hat{j}$. The charge moves in a magnetic field that is for $y > 0$ out of page and given by $B_1\hat{z}$ and for $y < 0$ into the page and given $-B_2\hat{z}$. The charge's subsequent trajectory is shown in the sketch. From this information, we can deduce that :



(A) $q > 0$ and $|B_1| < |B_2|$

(B) $q < 0$ and $|B_1| < |B_2|$

(C) $q > 0$ and $|B_1| > |B_2|$

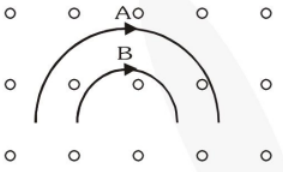
(D) $q < 0$ and $|B_1| > |B_2|$

Q23. A tightly-wound, long solenoid carries a current of 2.00 A. An electron is found to execute a uniform circular motion inside the solenoid with a frequency of 1.00×10^8 rev/s. Find the number of turns per metre in the solenoid.

(A) 500 Turns/m (B) 1020 Turns/m

(C) 1232 Turns/m (D) 1420 Turns/m

Q24. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively and the trajectories are as shown in the figure. Then :



(A) $m_A v_A < m_B v_B$

(B) $m_A v_A > m_B v_B$

(C) $m_A < m_B$ and $v_A < v_B$

(D) $m_A = m_B$ and $v_A = v_B$

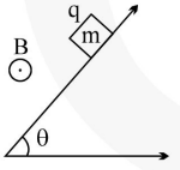
Q25. A block of mass m & charge q is released on a long smooth inclined plane magnetic field B is constant, uniform, horizontal and parallel to surface as shown. Find the time from start when block loses contact with the surface.

(A) $\frac{m\cos\theta}{qB}$

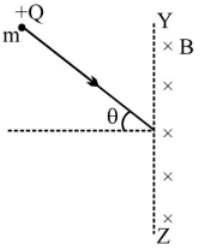
(B) $\frac{m\cos ec\,\theta}{qB}$

(C) $\frac{m\cot\theta}{qB}$

(D) none



Q26. A particle with charge +Q and mass m enters a magnetic field of magnitude B, existing only to the right of the boundary YZ. The direction of the motion of the particle is perpendicular to the direction of B. Let $T = 2\pi\frac{m}{QB}$. The time spent by the particle in the field will be :



(A) Tθ (B) 2Tθ

(C) $T\left(\frac{\pi+2\theta}{2\pi}\right)$ (D) $T\left(\frac{\pi-2\theta}{2\pi}\right)$

Q27. In the previous question, if the particle has –Q charge, the time spend by the particle in the field will be :

(A) Tθ (B) 2Tθ

(C) $T\left(\frac{\pi+2\theta}{2\pi}\right)$ (D) $T\left(\frac{\pi-2\theta}{2\pi}\right)$

Q28. A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from x = a to x = b. The minimum value of v required so that the particle can just enter the region x > b is -

(A) qbB/m (B) q(b – a)B/m

(C) qaB/m (D) q(b + a)B/2m

Q29. H^+, He^+ and O^{++} all having the same kinetic energy pass throuth a region in which is a uniform magnetic field perpendicular to their velocity. The masses of H^+, He^+ and O^{++} are 1 amu, 4 amu and 16 amu respectively then :-

(a) H^+ will be deflected most

(b) O^{++} will be deflected most

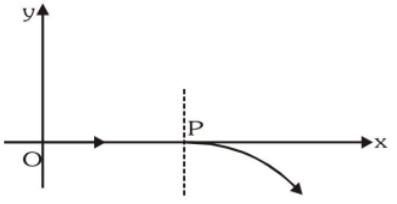
(c) He^+ and O^{++} will be deflected equally

(d) all will be deflected equally

(A) a, b (B) a, b, c

(C) only a (D) a, c

Q30. For a positively charged particle moving in a x – y plane initially along the x–axis, there is a sudden change in it path due to the presence of electric and/or magnetic field beyond P. The curved path is shown in the x – y plane and is found to be non–circular. Which one of the following combinations is possible?



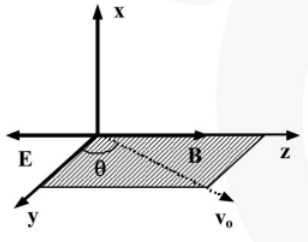
(A) $\vec{E}=0;\vec{B}=b\hat{j}+c\hat{k}$

(B) $\vec{E}=a\hat{i};\vec{B}=c\hat{k}+a\hat{i}$

(C) $\vec{E}=0;\vec{B}=c\hat{j}+b\hat{k}$

(D) $\vec{E}=a\hat{i};\vec{B}=c\hat{k}+b\hat{j}$

Q31. In a certain region uniform electric field E and magnetic field B are present in the opposite direction. At the instant t = 0, a particle of mass m carrying a charge q is given velocity v_o at an angle θ, with the y axis, in the yz plane. The time after which the speed of the particle would be minimum is equal to -



(A) $\frac{mv_o}{qE}$ (B) $\frac{mv_o\sin\theta}{qE}$

(C) $\frac{mv_o\cos\theta}{qE}$ (D) $\frac{2\pi m}{qB}$

Q32. There exists a uniform magnetic and electric field of magnitude 1 T and 1 V/m respectively along positive y–axis. A charged particle of mass 1 kg and of charge 1 C is having velocity 1 m/s along x–axis and is at origin at t = 0. Then the co–ordinates of particle at time π seconds will be -

(A) (0, 1, 2) (B) $(0, -\pi^2/2, -2)$

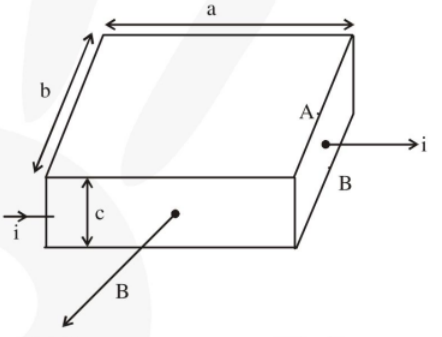
(C) $(2, \pi^2/2, 2)$ (D) $(0, \pi^2/2, 2)$

Q33. An electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction $\vec{B}$. If kinetic energy is doubled and magnetic induction tripled, the radius will become

(A) $\frac{3R}{2}$ (B) $\sqrt{\frac{3}{2}}R$

(C) $\sqrt{\frac{2}{9}}R$ (D) $\sqrt{\frac{4}{3}}R$

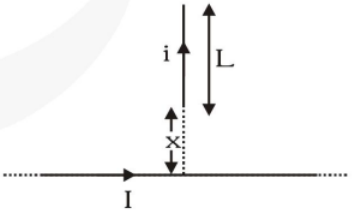
Q34. A current flows through a rectangular conductor in the presence of uniform magnetic field B pointing out of the page as shown. Then the potential difference $V_B - V_A$ is equal to (Assume charge carries in the conductor to be positively charged **moving with speed v**)



(A) Bvc (B) –Bvc

(C) Bvb (D) –Bvb

Q35. The magnetic force between wires as shown in figure is :-



(A) $\frac{\mu_0 iI^2}{2\pi}\ell n\left(\frac{x+\ell}{2x}\right)$

(B) $\frac{\mu_0 iI^2}{2\pi}\ell n\left(\frac{2x+\ell}{2x}\right)$

(C) $\frac{\mu_0 iI}{2\pi}\ell n\left(\frac{x+\ell}{x}\right)$

(D) None of these

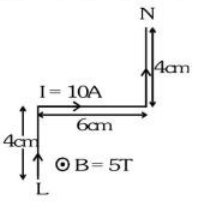
Q36. A wire 'LN' bent as shown in figure is placed in uniform perpendicular magnetic field of 5T. A 10A current flows through the wire. Magnetic force experienced by wire is :-

(A) 5 N

(B) 10 N

(C) 2.5 N

(D) 1.25 N



Q37. A solenoid 60 cm long and of radius 4.0 cm has 3 layers of windings of 300 turns each. A 2.0 cm long wire of mass 2.5 g lies inside the solenoid (near its centre) normal to its axis ; both the wire and the axis of the solenoid are in the horizontal plane. The wire is connected through two leads parallel to the axis of the solenoid to an external battery which supplies a current of 6.0 A in the wire. What value of current (with appropriate sense of circulation) in the windings of the solenoid can support the weight of the wire ? $g = 9.8\ ms^{-2}$.

(A) zero (B) 108 A

(C) 200 A (D) 458 A

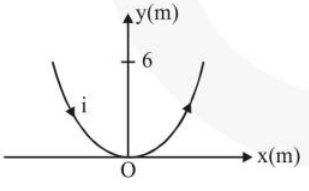
Q38. A wire is bent as a parabolic curve and kept in the x-y plane. The curve can be described by the equation $x^2 = 6y$. The wire carries a current i = 2A. If a uniform magnetic field $\vec{B} = 2\times10^{-3}\hat{k}$ tesla is applied, force experienced by the wire is nearly :-

(A) 1.5 $\hat{j}$N

(B) –1.5 $\hat{j}$N

(C) –0.05 $\hat{j}$N

(D) 2.5 $\hat{j}$N



Q39. A ring like metallic conductor of resistance R and radius a, caries a constant current I. The ratio of the angular momentum L of the conduction electrons (about the axis of the ring) and the magnetic field B at the centre of the ring satisfy [where e and m represent the magnitudes of the electronic charge and mass]

(A) $\frac{B}{L} \propto \frac{e^2}{m}$ (B) $\frac{B}{L} \propto e.m$

(C) $\frac{B}{L} \propto \frac{e}{m}$ (D) $\frac{B}{L} \propto \frac{m}{e}$

Q40. A charge of 3.14×10^{-6} C is distributed uniformly over a circular ring of radius 20.0 cm. The ring rotates about its axis with an angular velocity of 60.0 rad/s. Find the ratio of the electric field to the magnetic field at a point on the axis at a distance of 5.00 cm from the centre.

(A) zero (B) 0.5×10^{15} m/s

(C) 1.88×10^{15} m/s (D) 5.88×10^{15} m/s

Q41. Due to the flow of current in a circular loop of radius R, the magnetic field produced at the centre of the loop is B. The magnetic moment of the loop is :-

(A) $BR^3/2\ \pi\mu_0$ (B) $2\pi\ BR^3/\mu_0$

(C) $BR^2/2\pi\mu_0$ (D) $2\pi\ BR^2/\mu_0$

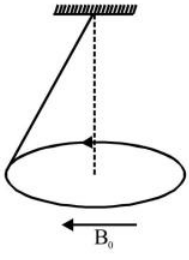
Q42. A uniform current carrying ring of mass m and radius R is connected by a massless string as shown. A uniform magnetic field B_0 exist in the region to keep the ring in horizontal position, then the current in the ring is (ℓ = length of string)

(A) $\frac{mg}{\pi RB_0}$

(B) $\frac{mg}{RB_0}$

(C) $\frac{mg}{3\pi RB_0}$

(D) $\frac{mg\ell}{\pi R^2 B_0}$



Q43. A solenoid of length 0.4 m, having 500 turns and 3A current flows through it. A coil of radius 0.01 m and have 10 turns and carries current of 0.4 A has to placed such that its axis is perpendicular to the axis of solenoid, then torque on coil will be:-

(A) 5.92×10^{-6} N.m

(B) 5.92×10^{-5} N.m

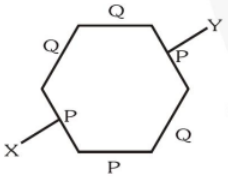
(C) 5.92×10^{-4} N.m

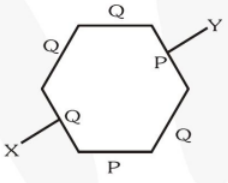
(D) 0.592×10^{-3} N.m

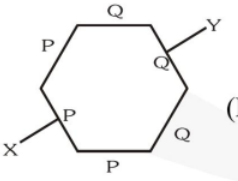
ONE OR MORE THAN ONE CORRECT TYPE

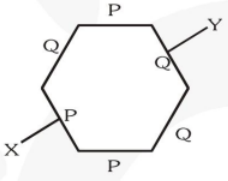
Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. In the following hexagons, made up of two different material P and Q, current enters and leaves from points X and Y respectively. In which case the magnetic field at its centre is not zero

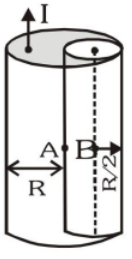
(A) 

(B) 

(C) 

(D) 

Q2. From a cylinder of radius R, a cylinder of radius R/2 is removed, as shown. Current flowing in the remaining cylinder is I. Magnetic field strength is -



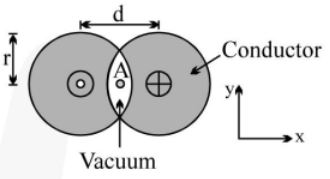
(A) zero at point A

(B) zero at point B

(C) $\frac{\mu_0 I}{3\pi R}$ at point A

(D) $\frac{\mu_0 I}{3\pi R}$ at point B

Q3. Two long conductors are arranged as shown above to form overlapping cylinders, each of radius r, whose centers are separated by a distance d. Current of density J flows into the plane of the page along the shaded part of one conductor and an equal current flows out of the plane of the page along the shaded portion of the other, as shown. What are the magnitude and direction of the magnetic field at point A?



(A) $(\mu_0/2\pi)\pi dJ$, in the +y-direction

(B) $(\mu_0/2\pi)d^2/r$, in the +y-direction

(C) $(\mu_0/2\pi)4d^2J/r$, in the –y-direction

(D) $(\mu_0/2\pi)Jr^2/d$, in the –y-direction

Q4. A charged particle is projected in a magnetic field $\vec{B} = 10\hat{k}$ T from the origin in x-y plane. The particle moves in a circle and just touches a straight line y = 5 (m) at point $\left(5\sqrt{3}, 5\right)$ (m). Then

(mass of particle = 5×10^{-5} kg, charge = 1 μC)

(A) The particle is projected at an angle 60° with x–axis

(B) The radius of the circle is 10 m

(C) The speed of the particle is 2m/s

(D) The speed of the particle is $5\sqrt{3}$ m/s

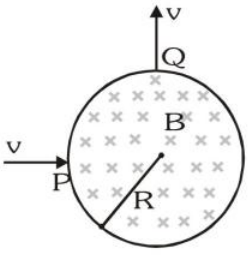
Q5. A particle of charge 'q' and mass 'm' enters normally (at point P) in a region of magnetic field with speed v. It comes out normally from Q after time T as shown in figure. The magnetic field B is present only in the region of radius R and is uniform. Initial and final velocities are along radial direction and they are perpendicular to each other. For this to happen, which of the following expression(s) is/are correct :-

(A) $B = \frac{mv}{qR}$

(B) $T = \frac{\pi R}{2v}$

(C) $T = \frac{\pi m}{2qB}$

(D) None of these



Q6. Which of the following statement is correct?

(A) A charge particle enters a region of uniform magnetic field at an angle 85° to magnetic lines of force. The path of the particle is a circle.

(B) An electron and a proton are moving with the same kinetic energy along the same direction. When they pass through uniform magnetic field perpendicular to their direction of motion, they describe circular path.

(C) There is no change in the energy of a charged particle moving in a magnetic field although magnetic force acts on it.

(D) Two electrons enter with the same speed but in opposite direction in a uniform transverse magnetic field. Then the two describe circle of the same radius and these move in the same direction.

Q7. A particle of mass m and charge q moving with velocity $\vec{v}$ enters a region of uniform magnetic field of induction $\vec{B}$. Then -

(A) its path in the region of the field is always circular

(B) its path in the region of the field is circular if $\vec{v}.\vec{B} = 0$

(C) its path in the region of the field is a straight line if $\vec{v} \times \vec{B} = 0$

(D) distance travelled by the particle in time T does not depend on the angle between $\vec{v}$ and $\vec{B}$

Q8. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a, b and c be the ratio of their time periods, radii and pitches of the helical paths than -

(A) abc =1 (B) abc >1

(C) abc <1 (D) a = bc

Q9. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ with velocity $\vec{v} = v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time

(A) $t = \frac{2mv_0}{qE_0}$ (B) $t = \frac{2Bq}{mv_0}$

(C) $t = \frac{\sqrt{3}\,Bq}{mv_0}$ (D) $t = \frac{\sqrt{3}\,mv_0}{qE_0}$

Q10. A particle of charge q and velocity v passes undeflected through a space with non–zero electric field E and magnetic field B. The undeflecting conditions will hold if

(A) signs of both q and E are reversed

(B) signs of both q and B are reversed

(C) both B and E are changed in magnitude, but keeping the product of |B| and |E| fixed.

(D) both B and E are doubled in magnitude

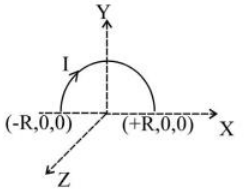
Q11. A semi circular current carrying wire having radius R is placed in x-y plane with its centre at origin 'O'. There is non-uniform magnetic field $\vec{B} = \frac{B_0 x}{2R}\hat{k}$ (here B_0 is +ve constant) is existing in the region. The magnetic force acting on semi circular wire will be along :

(A) – x-axis

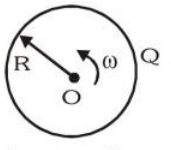
(B) + y-axis

(C) – y-axis

(D) + x-axis



Q12. A nonconducting disc having uniform positive charge Q, is rotating about its axis with uniform angular velocity ω. The magnetic field at the centre of the disc is–



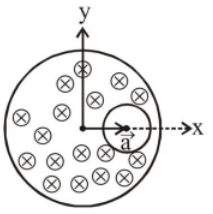
(A) directed outward

(B) having magnitude $\frac{\mu_0 Q\omega}{4\pi R}$

(C) directed inwards

(D) having magnitude $\frac{\mu_0 Q\omega}{2\pi R}$

Q13. Figure shows cross-section view of a infinite cylindrical wire with a cylinderical cavity, current density is uniform $\vec{J} = -j_0\hat{k}$ as shown in figure.



(A) Field inside cavity is uniform

(B) Field inside cavity is along $\vec{a}$

(C) Field inside cavity is perpendicular to $\vec{a}$

(D) If an electron is projected with velocity $v_0\hat{j}$ it will move undeviated before colliding with cavity wall.

Q14. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii a and b respectively. When a current I passes through the coil, the magnetic field at the centre is :

(A) $\frac{\mu_0 NI}{b}$ (B) $\frac{2\mu_0 NI}{a}$

(C) $\frac{\mu_0 NI}{2(b-a)}\ell n\frac{b}{a}$ (D) $\frac{\mu_0 I^N}{2(b-a)}\ell n\frac{b}{a}$

Q15. A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity $[u\hat{i} - v\hat{j}]$. Uniform electric, magnetic fields exist in the region along the +y direction, of magnitude E and B. The particle will definitely return to the origin once if

(A) $[vB/2\pi E]$ is an integer

(B) $(u^2 + v^2)^{1/2}\ [B/\pi E]$ is an integer

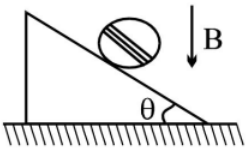
(C) $[vB/\pi E]$ is an integer

(D) $[uB/\pi E]$ is an integer

Q16. A metal ring of radius r = 0.5 m with its plane normal to a uniform magnetic field B of induction 0.2 T carries a current I = 100 A. The tension in newtons developed in the ring is:

(A) 100 (B) 50 (C) 25 (D) 10

Q17. In the figure shown a coil of single turn is wound on a sphere of radius R and mass m. The plane of the coil is parallel to the plane and lies in the equatorial plane of the sphere. Current in the coil is i. The value of B if the sphere is in equilibrium is -



(A) $\frac{mg\cos\theta}{\pi iR}$ (B) $\frac{mg}{\pi iR}$

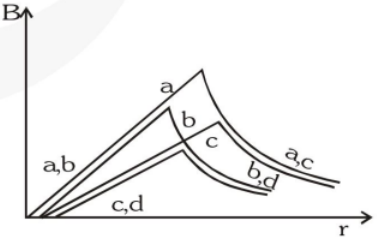
(C) $\frac{mg\tan\theta}{\pi iR}$ (D) $\frac{mg\sin\theta}{\pi iR}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Questions no. 18 to 20

Curves in the graph shown give, as functions of radial distance r, the magnitude B of the magnetic field inside and outside four long wires a, b, c and d, carrying currents that are uniformly distributed across the cross–sections of the wires. Overlapping portions of the plots are indicated by double labels.



Q18. Which wire has the greatest radius?

(A) a (B) b (C) c (D) d

Q19. Which wire has the greatest magnitude of the magnetic field on the surface?

(A) a (B) b (C) c (D) d

Q20. The current density in wire a is–

(A) greater than in wire c

(B) less than in wire c

(C) equal to that in wire c

(D) not comparable to that of in wire c due to lack of information

Paragraph for Question No. 21 to 24

In uniform magnetic field, if angle between $\vec{v}$ and $\vec{B}$ is $0° < \theta < 90°$, path of the particle is helix, with its axis parallel to $\vec{B}$ and plane perpendicular to $\vec{B}$. Here $\vec{v}$ is the velocity vector of the particle and $\vec{B}$ magnetic field vector. Let v_1 be the component of $\vec{v}$ along $\vec{B}$ and v_2 the component perpendicular to $\vec{B}$. Suppose p is the pitch, T the time period and r the radius of helix.

Then : $T = \frac{2\pi m}{Bq}$, $r = \frac{mv_2}{Bq}$ and $p = (v_1)\,(T)$.

A charged particle (q, m) is released from origin with velocity $\vec{v} = v_0\hat{i}$ in a uniform magnetic field $\vec{B} = \frac{B_0}{2}\hat{i} + \frac{\sqrt{3}B_0}{2}\hat{j}$.

Q21. Pitch of the helical path described by the particle is :

(A) $\frac{2\pi m v_0}{B_0 q}$

(B) $\frac{\sqrt{3}\pi m v_0}{2B_0 q}$

(C) $\frac{\pi m v_0}{B_0 q}$

(D) $\frac{2\sqrt{3}\pi m v_0}{B_0 q}$

Q22. z–component of velocity is $\frac{\sqrt{3}v_0}{2}$ after time t = ...

(A) $\frac{2\pi m}{B_0 q}$ (B) $\frac{\pi m}{B_0 q}$

(C) $\frac{\pi m}{2B_0 q}$ (D) $\frac{2\pi m}{4B_0 q}$

Q23. Maximum z–coordinate of the particle is :

(A) $\frac{\sqrt{3}mv_0}{B_0 q}$ (B) $\frac{2\sqrt{3}mv_0}{B_0 q}$

(C) $\frac{2mv_0}{B_0 q}$ (D) $\frac{mv_0}{B_0 q}$

Q24. When z–coordinate has its maximum value :

(A) $v_x = 0$

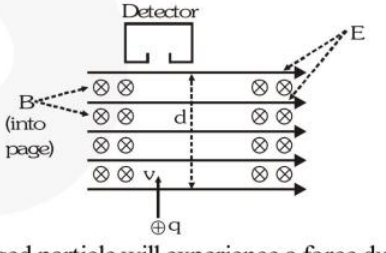
(B) $v_y = 0$

(C) Both (A) and (B) are correct

(D) Both (A) and (B) are wrong

Paragraph for Questions no. 25 to 28

A velocity filter uses the properties of electric and magnetic fields to select charged particles that are moving with a specific velocity. Charged particles with varying speeds are directed into the filter as shown in figure. The filter consists of an electric field E and a magnetic field B, each of constant magnitude, directed perpendicular to each other as shown. The particles that move straight through the filter with their direction unaltered by the fields have the specific filter speed, v_0. Those with speeds to v_0 may experience sufficiently little deflection that they also enter the detector.



The charged particle will experience a force due to the electric field given by the relationship $\vec{F} = q\vec{E}$, where q is the charge of the particle and $\vec{E}$ is the electric field. The moving particle will also experience a force due to the magnetic field. This force acts to oppose the force due to the electric field. The strength of the force due to the magnetic field is given by the relationship $\vec{F} = q(\vec{v} \times \vec{B})$, where q is the charge of the particle, $\vec{v}$ is the speed of the particle, and $\vec{B}$ is the magnetic field strength. When the forces due to the two fields are equal and opposite, the net force on the particle will be zero, and the particle will pass through the filter with its path unaltered. The electric and magnetic field strengths can be adjusted to choose the specific velocity to be filtered.

All mic. The effects of gravity can be neglected.

Q25. The electric and magnetic fields in the filter of figure are adjusted to detect particles with positive charge q of a certain speed, v_0. Which of the following expressions is equal to this speed ?

(A) $B/(q^2E)$ (B) $E/(q^2B)$

(C) B/E (D) E/B

Q26. Which of the following is true about the velocity filter shown in figure ?

(A) It would not work with negatively charged particles.

(B) The wider the detector entrance, the more narrow the range of speed detected.

(C) The greater the distance d, the more narrow the range of speeds detected.

(D) The detector may not detect a charged particle with the desired filter speed if its charge is too high.

Q27. Which of the following statements is true regarding a charged particle that is moving through the filter at a speed that is less than the filter speed ?

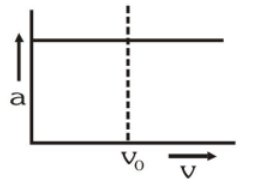
(A) It experiences a greater force due to the magnetic field than due to the electric field.

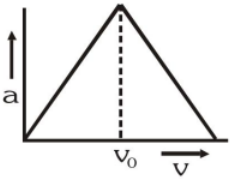
(B) It experiences a greater force due to the electric field than due to the magnetic field.

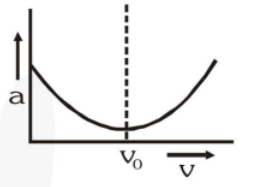
(C) It experiences equal force due to both fields but greater acceleration due to the electric field.

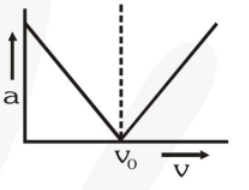
(D) It experiences equal force due to both fields but greater acceleration due to the magnetic field.

Q28. Particles of identical mass and charge are sent through the filter at varying speeds, and the magnitude of acceleration of each particle is recorded as it first begins to be deflected. If the filter is set to detect particles of speed v_0, which one of the following is correct graph between acceleration and velocity of particle:

(A) 

(B) 

(C) 

(D) 

Paragraph for Question No. 29 to 32

There are two infinite parallel current carrying wire in vertical plane. Lower wire is fixed and upper wire is having a linear mass density λ. Two wires are carrying current I_1 and I_2. Now upper wire is placed in a magnetic field produced by lower wire. Magnetic field due to lower wire at the location of upper wire is $\frac{\mu_0 I}{2\pi d}$ where $I_1 \rightarrow$ current in lower wire, $d \rightarrow$ separation between wire. Force on any small portion of upper wire having length $d\ell$ is $dF = \frac{\mu_0 I_1 I_2 d\ell}{2\pi d}$ where $I_2 \rightarrow$ current in the upper wire. If directions of current in the wires are appropriate then upper wire can be in equilibrium if its weight is balanced by magnetic force. Now answer the following questions

Q29. Equilibrium separation between the two wires is

(A) $\frac{\mu_0 I_1 I_2}{4\pi\lambda g}$ (B) $\frac{\mu_0 I_1 I_2}{2\pi\lambda g}$

(C) $\frac{\mu_0 I}{\pi\lambda g}$ (D) None of these

Q30. The upper wire can be in equilibrium if
(A) Direction of current in both wires is same
(B) Direction of current in both wires is opposite
(C) Equilibrium does not depend upon the direction of currents
(D) None of these

Q31. If upper wire is slightly displaced from its mean position and released, it will perform simple harmonic motion. As wire moves the total mechanical energy of wire.
(A) Remains constant
(B) Changes
(C) We can't say anything about mechanical energy in magnetic field
(D) None of the above

Q32. Consider wire at lower extreme position and upper extreme position. Kinetic energy of wire is zero at both the position then -
(A) Gravitational potential energy is also same at both positions
(B) Gravitation potential energy is different at both positions
(C) Gravitational potential energy change can be neglected because displacement of wire from the mean position is very small
(D) None of the above

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q33. A charged particle with some initial velocity is projected in a region where uniform electric and/or magnetic fields are present. In Column-I information about the existence of electric and/or magnetic field and direction of initial velocity of charged particle are given, while in column II the possible paths of charged particle is mentioned. Match the entries of Column I with the entries of Column II.

Column–I	**Column–II**
(A) $\vec{E}=0, \vec{B}\neq 0$ and initial velocity is at an unknown angle with $\vec{B}$	(p) Straight line
(B) $\vec{E}\neq 0, \vec{B}=0$ and initial velocity is at an unknown angle with $\vec{E}$	(q) Parabola
(C) $\vec{E}\neq 0, \vec{B}\neq 0, \vec{E}\parallel\vec{B}$ and initial velocity is perpendicular to $\vec{E}$	(r) Circular
(D) $\vec{E}\neq\vec{B}, B\neq 0, \vec{E}$ perpendicular $\vec{B}$ and initial velocity is perpendicular to both $\vec{E}$ and $\vec{B}$	(s) Helical path with nonuniform pitch (t) Helical path with uniform pitch

Q34. **Column I** (Magnetic moment of) **Column II**

(A) a uniformly charged ring rotating uniformly about its axis (p) $\frac{q\omega r^2}{5}$

(B) a charged particle rotating uniformly about a point (q) $\frac{q\omega r^2}{4}$

(C) a uniformly charged disk rotating uniformly about its axis (r) $\frac{q\omega r^2}{3}$

(D) a uniformly charged spherical shell rotating uniformly about one of its diameter (s) $\frac{q\omega r^2}{2}$

(E) a uniformly charged sphere rotating uniformly about one of its diameter (t) $q\omega r^2$

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	B	B	B	B	B	D	A	A	C
11	12	13	14	15	16	17	18	19	20
B	D	D	D	C	A	B	C	B	A
21	22	23	24	25	26	27	28	29	30
C	C	C	A	B	C	D	B	A	A
31	32	33							
C	A	D							

Ex-1A

1. B,C **2.** B,D **3.** A,B,C **4.** A,B,C **5.** A,B,C

6. A,B **7.** A,B,C **8.** C **9.** A

10. A → q,s ; B → p,s ; C → q,s ; D → p,s

11. A → r ; B → q ; C → s

12. A → q ; B → p,q ; C → p,q,r

13. 1.2 cm, 4.35 cm **14.** (a) $\frac{\mu_0 I}{4R}\otimes$ (b) $\frac{\mu_0 I}{8R}\otimes$

15. $\mu_0 nI$ **16.** $\sqrt{2}M$

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	B	D	D	A	C	B	D	A	C
11	12	13	14	15	16	17	18	19	20
D	C	C	D	B	C	A	A	A	C
21	22	23	24	25	26	27	28	29	30
D	A	D	B	C	C	D	B	D	B
31	32	33	34	35	36	37	38	39	40
B	D	C	A	C	A	B	C	C	C
41	42	43							
B	A	A							

Ex-2A

1. A,B **2.** C,D **3.** A **4.** A **5.** A,B,C
6. B,C **7.** B,C,D **8.** A,D **9.** D **10.** D
11. A **12.** A,D **13.** A,C,D **14.** C **15.** C
16. D **17.** B **18.** C **19.** A **20.** A
21. C **22.** C **23.** A **24.** D **25.** D
26. C **27.** B **28.** D **29.** B **30.** B
31. B **32.** B
33. (A) $\rightarrow$ (p,r, t) ; (B) $\rightarrow$ (p,q) ; (C) $\rightarrow$ (s) ; (D) $\rightarrow$ (p)
34. A $\rightarrow$ s ; B $\rightarrow$ s ; C $\rightarrow$ q ; D $\rightarrow$ r ; E $\rightarrow$ p

-: Important Notes & Formulas :-